


# SF1811 Optimization

## Lecture 2: Basic feasible solutions

Chapter 4

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# Basic feasible solutions

Main question:

Why are basic feasible solutions the relevant candidates for optimality?

Today:

- Basic and nonbasic variables.
- Basic solutions.
- Basic feasible solutions.
- Basic feasible solutions and vertices.
- The fundamental theorem of linear programming.

# Linear programming in standard form

Consider a linear program in standard form:

$$\begin{aligned} \text{minimize} \quad & c^{\mathsf{T}} x \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0, \end{aligned}$$

where

$$A \in \mathbb{R}^{m \times n}, \qquad b \in \mathbb{R}^{m}, \qquad c, x \in \mathbb{R}^{n}.$$

Assume that

$$\operatorname{rank}(A) = m, \qquad m < n.$$

The $m$ equations do not determine all $n$ variables uniquely.

# Choosing the nonbasic variables

Let $N = (N_1, \ldots, N_{n-m})$ denote a tuple of $n - m$ of indices. The corresponding variables, collected in $x_N$, are set equal to zero:

$$x_N = 0.$$

Let $B = (B_1, \ldots, B_m)$ denote a tuple of the remaining indices, and let $x_B$ denote the vector of corresponding variable values.

Correspondingly,

$$A = \begin{pmatrix} A_B & A_N \end{pmatrix}, \qquad Ax = A_B x_B + A_N x_N.$$

As $x_N = 0$, the remaining variables $x_B$ must satisfy

$$A_B x_B = b.$$

If $A_B$ is nonsingular, $B$ is called a basis and $x_B$ are the basic variables.

We choose the nonbasic variables. The equations determine the basic variables.

# Basis and basic solution

Let $B$ denote a basis and let $N$ denote the tuple of nonbasic indices.
The corresponding columns are collected in

$$A_B = \begin{pmatrix} a_{B_1} & \cdots & a_{B_m} \end{pmatrix}, \qquad A_N = \begin{pmatrix} a_{N_1} & \cdots & a_{N_{n-m}} \end{pmatrix}.$$

The solution defined by

$$x_N = 0, \qquad A_B x_B = b,$$

is the basic solution associated with the basis $B$.

# Basic solution and basic feasible solution

A basic solution is defined by

$$x_N = 0, \qquad A_B x_B = b.$$

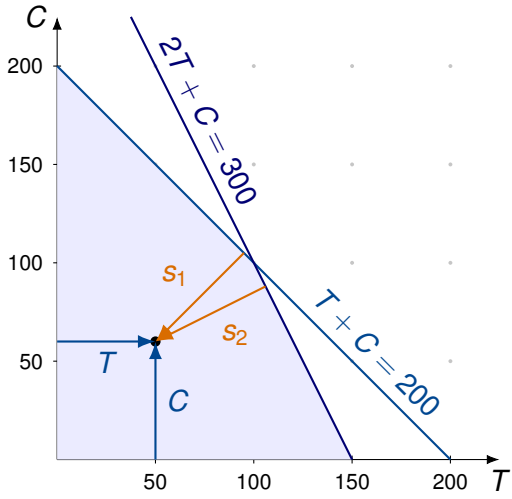
It is a basic feasible solution if

$$x_B \geq 0.$$

| Step | Operation | Question |
|---|---|---|
| Choose | Set $x_N = 0$ | Does $A_B$ form a basis? |
| Solve | Solve $A_B x_B = b$ | What are the basic-variable values? |
| Check | Inspect $x_B$ | Is $x_B \geq 0$? |

Constructing a basic solution is relatively easy. Whether the resulting solution is feasible is known only after solving $A_B x_B = b$.

# The Tables and chairs problem



# Tables and chairs: a basic feasible solution

The standard-form constraints are

$$\begin{aligned} T + C + s_1 &= 200, \\ 2T + C + s_2 &= 300, \end{aligned} \qquad T, C, s_1, s_2 \geq 0.$$

Here $m = 2$ and $n = 4$. We therefore choose two nonbasic variables.

Choose

$$T = C = 0.$$

# Tables and chairs: another basic feasible solution

The standard-form constraints are

$$T + C + s_1 = 200,$$
$$2T + C + s_2 = 300.$$

Choose

$$s_1 = s_2 = 0.$$

# Tables and chairs: a basic solution that is infeasible

The standard-form constraints are

$$T + C + s_1 = 200,$$
$$2T + C + s_2 = 300.$$

Choose

$$T = s_2 = 0.$$

# Basic feasible solutions and vertices

For the Tables and chairs problem:

| Nonbasic variables | Basic-variable values | Vertex |
|---|---|---|
| $T, C$ | | |
| $s_1, s_2$ | | |
| $C, s_1$ | infeasible | – |
| $C, s_2$ | | |
| $T, s_1$ | | |
| $T, s_2$ | infeasible | – |

The basic feasible solutions obtained algebraically correspond to the vertices found geometrically.

# The fundamental theorem of linear programming

**Theorem (The fundamental theorem of linear programming)**

*Consider a linear program in standard form.*

1. *If the linear program has a feasible solution, then it has a basic feasible solution.*
2. *If the linear program has an optimal solution, then it has an optimal basic feasible solution.*

Consequently, it is sufficient to search for an optimal solution among the basic feasible solutions.

# Proof idea: feasibility

- Start from a feasible solution $x$.
- Let $J = \{j : x_j > 0\}$.
  If the columns $a_j$, $j \in J$, are linearly dependent, there is a nonzero vector $d$ such that $Ad = 0$, $d_j = 0$, $j \notin J$.
  After replacing $d$ by $-d$, if necessary, we may assume that $d$ has at least one negative component.
- Hence, $A(x + td) = b$ for every $t$.
- Choose $t = \min_{j:d_j<0} \frac{x_j}{-d_j}$. Then $x + td \geq 0$, and at least one previously positive component becomes zero.
- Repeat until the columns corresponding to the remaining positive components are linearly independent. If necessary, complete these columns to a basis using linearly independent columns corresponding to zero variables.

# Proof idea: optimality

- Now suppose that the starting solution $x$ is optimal.
- Find $d$ as on the previous slide.
- For sufficiently small $t > 0$, both $x + td$ and $x - td$ are feasible.
- Their objective values are

$$c^{\mathsf{T}}(x + td) = c^{\mathsf{T}}x + t\,c^{\mathsf{T}}d, \quad \text{and} \quad c^{\mathsf{T}}(x - td) = c^{\mathsf{T}}x - t\,c^{\mathsf{T}}d.$$

- If $c^{\mathsf{T}}d \neq 0$, one of the two directions improves the objective value. This contradicts optimality. Therefore,

$$c^{\mathsf{T}}d = 0.$$

- The reduction can consequently be made without changing the objective value. Repeat as on the previous slide.

# Why not enumerate all bases?

There are at most

$$\binom{n}{m}$$

ways to choose $m$ columns from an $m \times n$ matrix.

Some choices do not form a basis, and some basic solutions are infeasible. Nevertheless, the number of possible bases can be very large.

The simplex method searches systematically among basic feasible solutions rather than enumerating all bases.

# Main ideas

After this lecture, you should be able to:

- explain the roles of $B$, $N$, $A_B$, $x_B$, and $x_N$;
- construct a basic solution by choosing $x_N = 0$;
- determine whether a basic solution is feasible;
- relate basic feasible solutions to vertices;
- explain why an optimal solution can be sought among basic feasible solutions.

Next lecture:

How does the simplex method move from
one basic feasible solution to a better one?